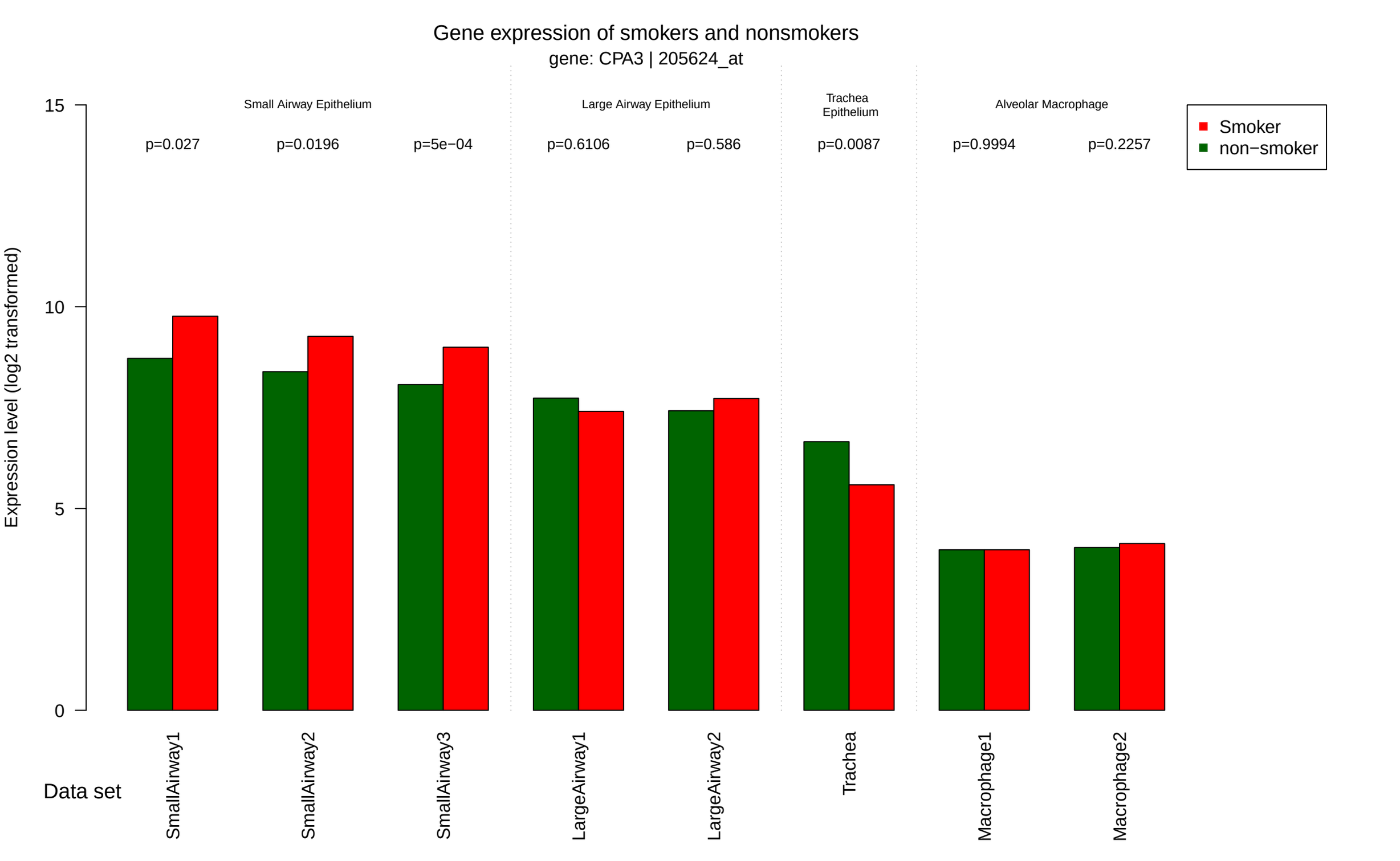
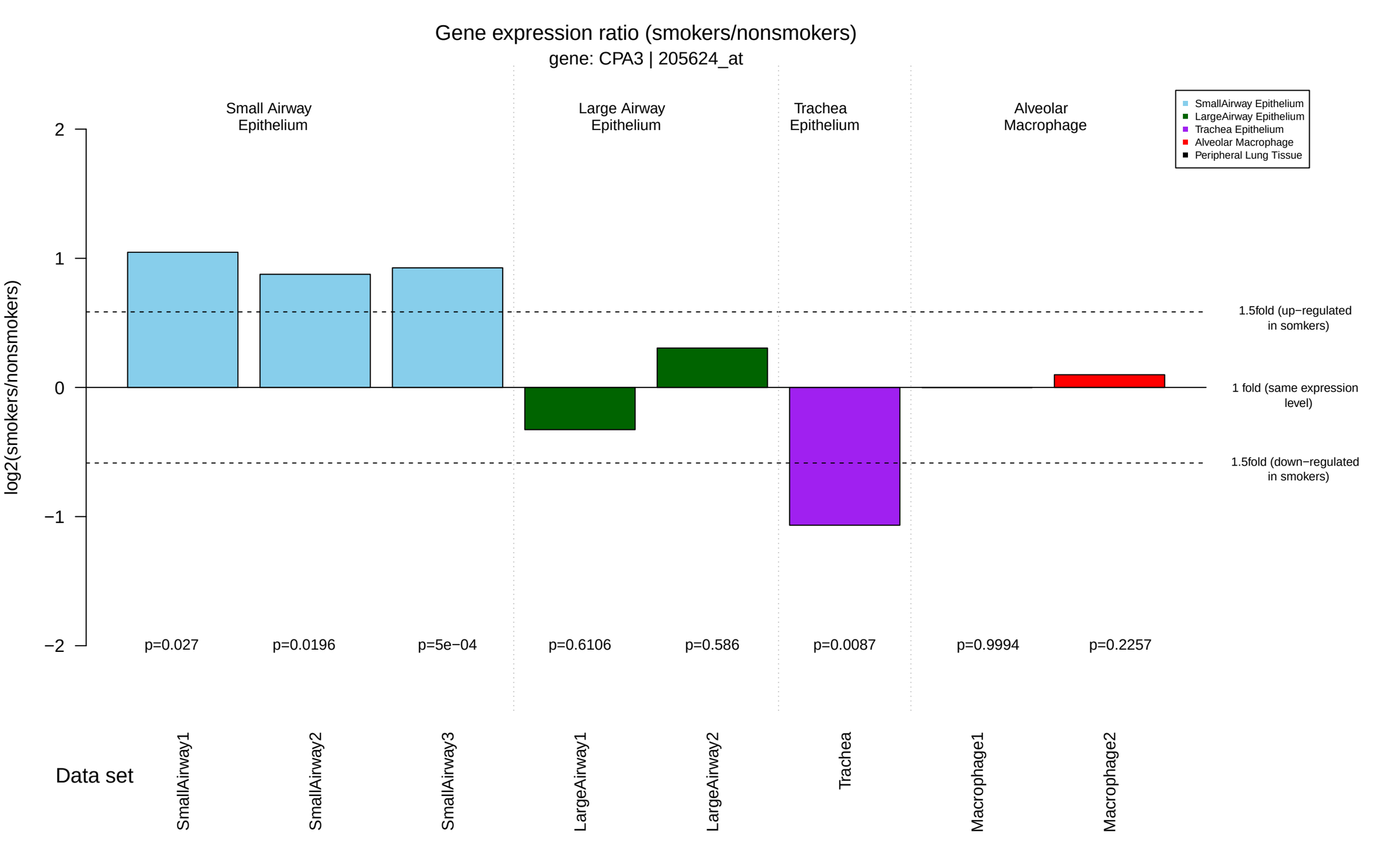
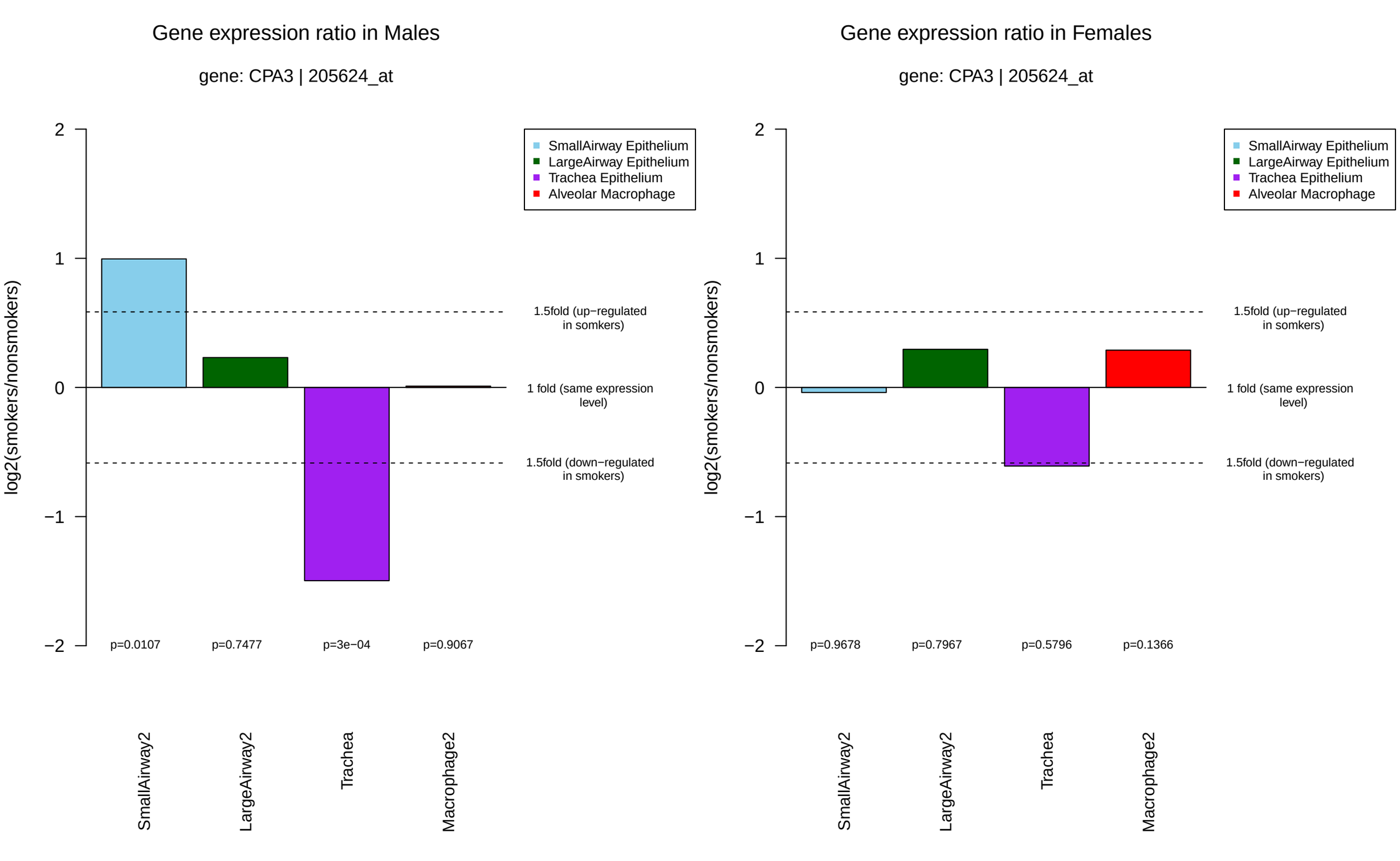
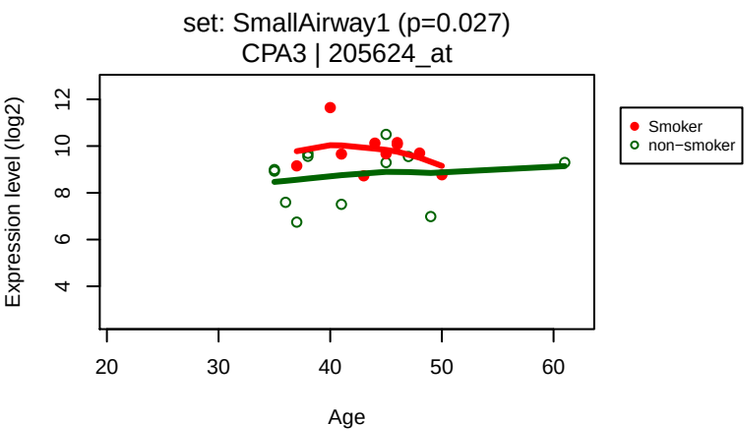
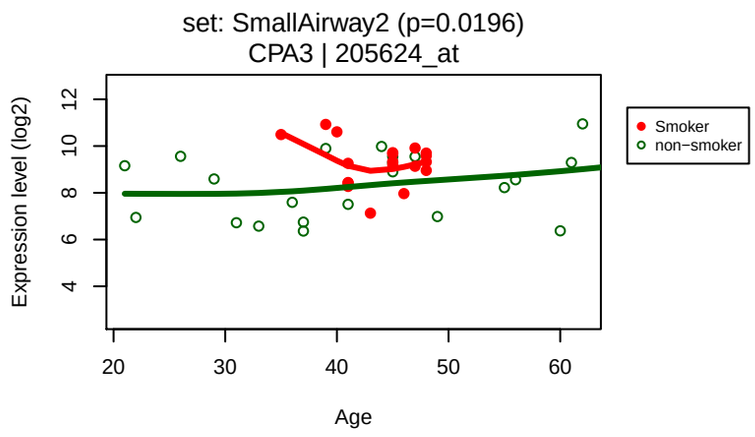
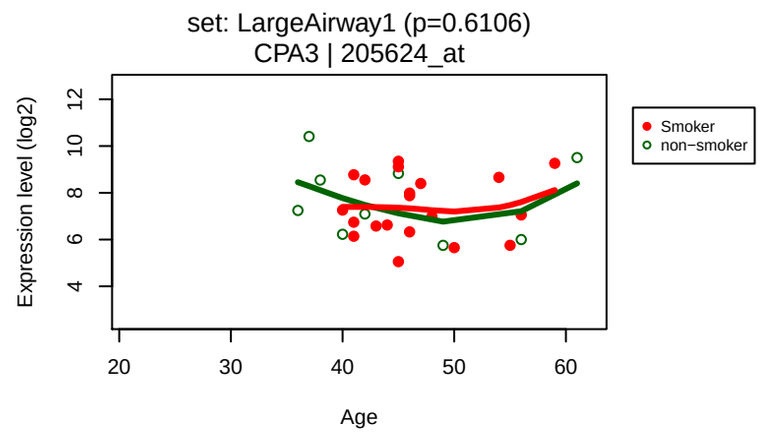

Gene expression of smokers and nonsmokers
gene: CPA3 | 205624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.027
p=0.0196
p=5e−04
p=0.6106
p=0.586
p=0.0087
p=0.9994
p=0.2257
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

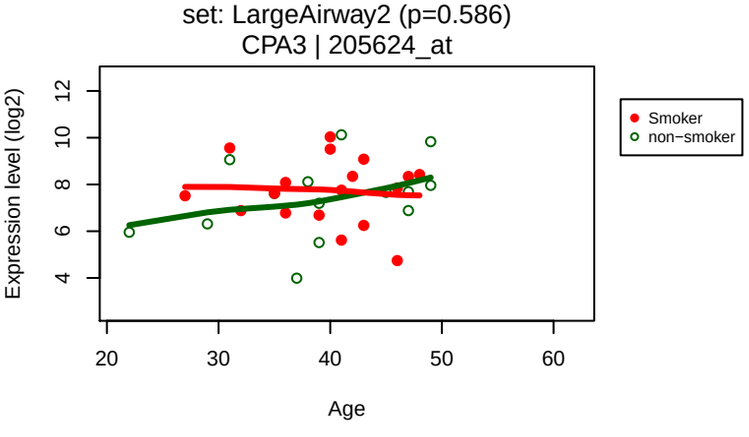

Gene expression ratio (smokers/nonsmokers)
gene: CPA3 | 205624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.027
p=0.0196
p=5e−04
p=0.6106
p=0.586
p=0.0087
p=0.9994
p=0.2257
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

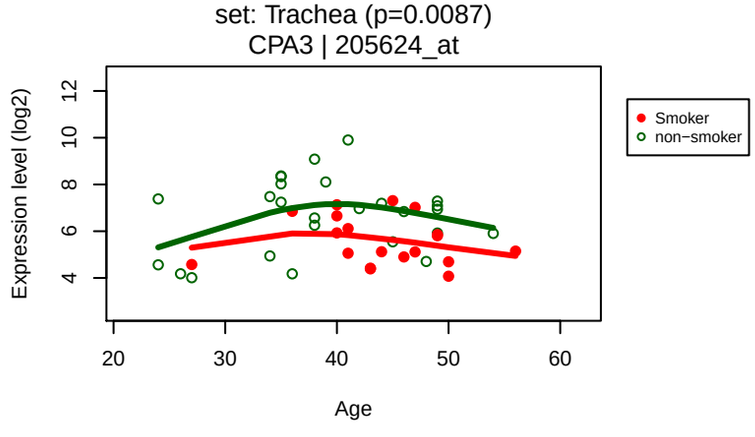

Gene expression ratio in Males
gene: CPA3 | 205624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.7477
p=3e−04
p=0.9067
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CPA3 | 205624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9678
p=0.7967
p=0.5796
p=0.1366
SmallAirway2
LargeAirway2
Trachea
Macrophage2

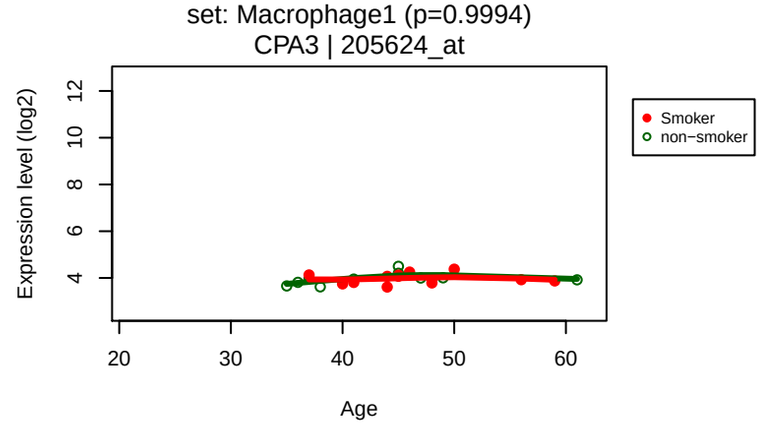

set: SmallAirway1 (p=0.027)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker

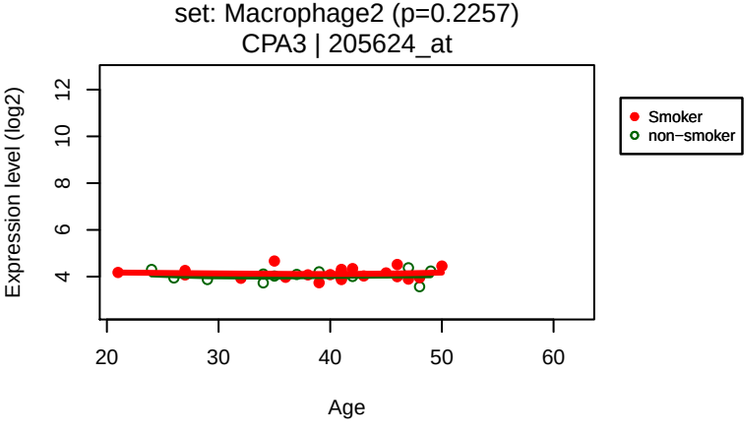

set: SmallAirway2 (p=0.0196)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker

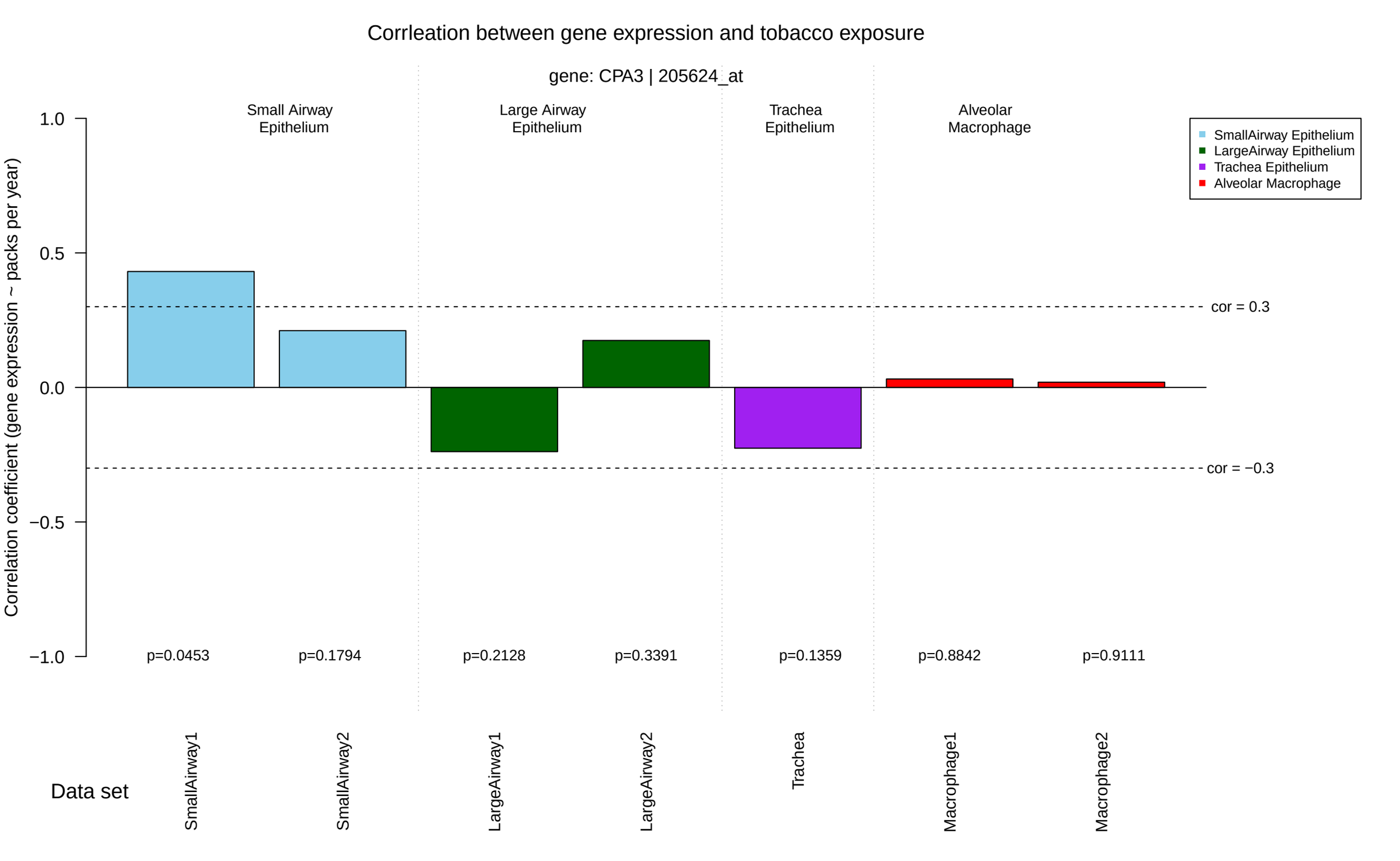

set: LargeAirway1 (p=0.6106)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.586)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0087)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9994)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2257)
CPA3 | 205624_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CPA3 | 205624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0453
p=0.1794
p=0.2128
p=0.3391
p=0.1359
p=0.8842
p=0.9111
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2